

ANDREAS MENTZELOPOULOS

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EDUCATION

Massachusetts Institute of Technology – MIT	2020 – 2025
PhD Mechanical Engineering and Computation, minor in Finance	
SM Mechanical Engineering	
University of Michigan – Ann Arbor	2016 – 2020
BSE Mechanical Engineering, minor in Mathematics	
BES Naval Architecture & Marine Engineering	

INTERESTS & BIO

Generative Modeling, Time-series Forecasting, Explainable AI, Deep Learning

I am an impact driven professional with a strong record of developing scalable machine-learning solutions. My expertise spans mathematical modeling, machine learning, optimization, probability, stochastic processes, and numerical methods. At Wayfair, I develop multi-touch attribution models that quantify the impact of marketing campaigns via explainable deep learning (Shapley), guiding the allocation of over \$1B in marketing spend. At MIT, I lead LOBSTgER, develop digital twins for marine structures, and run deep-virtual towing tanks! My previous experience includes roles in quant-trading, software, and engineering.

SKILLS

Computer languages	Python*, MATLAB*, C++, Julia	*expert
Machine/Deep Learning frameworks	PyTorch*, scikit-learn*, TensorFlow	
Generative-AI	Diffusion*, VAE*, GANs	
Time-series forecasting	Transformers*, LSTM*, GRU, XGBoost, LightGBM	

WORK EXPERIENCE

Wayfair, Marketing & Measurement Science, Machine Learning Scientist, Boston, MA 10/2025 – pres.

- 1. Developing marketing multi-touch attribution models via explainable deep learning.
 - a. Transitioning internal models to deep learning increasing flexibility and accuracy.
 - b. Developing Shapley-based pipelines for credit attribution which are validated against lift studies.

CITIC Securities CLSA, Quantitative Trading Strategies, Intern, New York, NY 06 – 08/2024

- 1. Developed the firm’s deep learning statistical arbitrage trading framework (conceptualization and coding).
 - a. Formulated market neutral long-short strategies.
 - b. Implemented a transformer-based stock returns forecasting model based on proprietary indicators.
 - c. Backtested the framework on the Hong Kong markets (public equities).
 - d. Outperformed internal strategies in terms of returns, Sharpe ratio, and portfolio turnover.

MathWorks, Engineering Development Group, Intern, Natick, MA 06 – 08/2023

- 1. Led quality engineering efforts for the Simscape family of products.
 - a. Designed and implemented test suites for simscape multibody example models, ensuring robustness and reliability across updates.
 - b. Implemented test suites for the Gas, Moist Air, and Isothermal Liquid libraries, assuring reliability and performance.
- 2. Re-designed and optimized the hydraulics of the customer-facing forklift example model, which was publicly released with MATLAB R2024a.

American Bureau of Shipping, Offshore Equipment Group, Intern, Houston, TX 06 – 08/2020

- 1. Conducted engineering reviews for pressure vessel designs in compliance with ASME Section VIII Div. 1 safety regulations for high-stakes offshore and marine environments.
- 2. Worked on the classification of BP’s Mad Dog Phase 2 – Argos semi-submersible offshore drilling platform, reviewing over 340 technical drawings.

3. Developed and verified allowable chemical cargo lists for 12 chemical tankers in accordance with the IBC Code, effective January 2021.

RESEARCH EXPERIENCE

Tow Tank | Stochastic Analysis and Nonlinear Dynamics Lab | MIT

10/2025-pres.

Research Affiliate

Graduate Student Research Assistant

09/2020 – 10/2025

1. LOBSTgER (Learning Oceanic Bioecological Systems Through generative Representations):

- Led the development of diffusion-based generative models to synthesize and enhance high-resolution underwater awareness photography images. In collaboration with photographer [Keith Ellenbogen](#), who provides training data.
- Developed unconditional latent diffusion models generating 640x1024 photography-like image; samples [here](#).
- Developed conditional restoration pipelines addressing low-visibility, color distortion, optical occlusion, and other challenges of underwater photography; samples [here](#).
- Recognized by [MIT News](#) (06/25) and featured at the Museum of Science in Boston (“Meet a Scientist Series”, 10/24; “Computer Science Education Weekend”, 12/24).
- Featured across: [MIT Sea Grant](#), [NOAA Sea Grant](#), MIT School of Engineering (X), MIT MechE (Instagram, LinkedIn, X), and highlighted on the Face of MIT MechE website.

2. Firstling Digital Twin:

- Developed a transformer-based deep neural network for underwater pipeline vibrations forecasting.
- High-accuracy predictions from sparse measurements accommodating wave inputs and other auxiliary data.
- Validated on data collected from BP’s Atlantis platform operating in the Gulf of America.

3. Virtual Towing Tank:

- Conducted studies to determine the most suitable generative model for pipe vibration data, comparing GANs, VAEs, and Diffusion models.
- Developed conditional latent diffusion models that generate densely sampled vibrations time-series based on physical parameters like tension and flow velocity as conditioning information, equivalent to altering experimental conditions in the lab.
- Validated on data collected at the MIT Towing Tank.
- Generated data are higher fidelity than CFD data and are obtained much faster with significantly less compute.

SELECTED PUBLICATIONS (full list: [google scholar](#), pre-prints [here](#))

1. **(Submitted) Mentzelopoulos, A.,** Fan, D., Sapsis, T., Karniadakis, G., Triantafyllou, MS, “[Deep generative modeling of vortex-induced vibrations](#)” (pre-print), Engineering Applications of Artificial Intelligence, 2025.
2. **Mentzelopoulos, A.,** Fan, D., Sapsis, T., Triantafyllou, MS, “[Deep learning vortex-induced vibrations: time-space forecasting with transformers](#)”, Journal of Fluids and Structures, 2025.
3. **Mentzelopoulos, A.,** Fan, D., Sapsis, T., Triantafyllou, MS, “[Variational autoencoders and transformers for multivariate time-series generative modeling and forecasting: Applications to vortex-induced vibrations](#)”, Ocean Engineering, 2024.
4. **Mentzelopoulos, A.** Ferrandis, J.d.A., Rudy, S., Sapsis, T., Triantafyllou, M.S, Fan, D., “[Data driven prediction and study of vortex induced vibrations by leveraging hydrodynamic coefficient databases learned from sparse sensors](#)”, Ocean Engineering, 2022.

SELECTED HONORS & AWARDS (for full list reference [personal website](#))

1. **Onassis Foundation Scholarship**, Onassis Foundation (most prestigious graduate scholarship awarded to Greek students globally) 12/2022
2. **MathWorks Fellowship**, MathWorks (awarded to 20 MIT students) 09/2022
3. **Society of Naval Architecture and Marine Engineering Award**, MIT (awarded to 1 student at MIT) 05/2021
4. **William M. Kennedy Scholarship**, Society of Naval Architects & Marine Engineers (awarded to 1 student globally) 04/2021
5. **MIT SMA2 Fellowship**, MIT (awarded to 1 graduate student at MIT) 09/2020
6. **Undergraduate Scholarship**, Society of Naval Architects & Marine Engineers (awarded to 4 students globally) 07/2019
7. **ABS Scholarship**, American Bureau of Shipping 05/2018

RELEVANT COURSEWORK (MIT)

Machine Learning

1. Machine Learning (6.7900)
2. Deep Learning (6.S898)
3. Parallel Computing & Scientific Machine Learning (6.7320)
4. Computer Vision (6.8300)
5. Nonlinear Optimization (6.7220)
6. Intro to Machine Learning (6.036)

Finance

1. Advanced Data Analytics and Machine Learning in Finance (Natural language processing with applications in Finance - 15.S08)
2. Financial Engineering (15.456)
3. Managerial Finance (15.041)

Mathematics

1. Numerical Methods for Partial Differential Equations (6.7330)
2. Stochastic Systems (2.122)
3. Numerical Fluid Mechanics (2.29)
4. Dynamics (2.032)
5. Marine Hydrodynamics (2.20)